PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Safety and efficacy of inpatient pulmonary rehabilitation for patients hospitalized with an acute exacerbation of chronic obstructive pulmonary disease: a systematic review protocol
AUTHORS	Zhu, Kai; Gill, Jagdeep; Kirkham, Ashley; Chen, Joel; Ellis, Amy; Crosbie, Stephanie; Denson-Camp, Holly; Peters, Hannah; Camp , Pat

VERSION 1 – REVIEW

REVIEWER	Chaplin, Emma University Hospitals of Leicester NHS Trust, Pulmonary Rehabilitation
REVIEW RETURNED	22-Feb-2021

GENERAL COMMENTS	This is an interesting systematic review, although I'm not sure how different it may be from the Puhan review as a lot of these studies had some elements of inpatient PR which you acknowledge are included in your inclusion criteria of types of interventions. However there are possibly more studies as this was done some time ago. In the Methods and Analysis; types of interventions - rehabilitation programmes may include self-management, education, dietary advice etc, how was this measured? The description of the time points was also slightly confusing as to the inclusion/ exclusion of studies. Data analysis describes the QoL and exercise capacity measures however this paper was to review the safety of inpatient PR as well as efficacy, and I was unable to find any description of how adverse events, readmission or LOS were analysed which are important in measuring the safety aspect. The authors may want to review the MID for the ISWT as this has now changed to 40m (Evans RA, Singh SJ. Minimum important difference of the incremental shuttle walk test distance in patients with COPD. Thorax 2019;74: 994-995)

REVIEWER	Jones, Arwel Monash University, Department of Allergy, Immunology and Respiratory Medicine
REVIEW RETURNED	09-Apr-2021

GENERAL COMMENTS	Thanks for the opportunity to review this manuscript. Pulmonary rehabilitation (PR) is one of the most effective interventions for people living with COPD. There are, however,

	some uncertainties on how best to deliver this intervention during and following acute exacerbations of COPD (AECOPD). The 2017 ATS/ERS guideline for management of AECOPD recommended the initiation of PR following discharge and not during the admission. However, this recommendation was limited by inconsistency in the evidence and the limited number of trials. Now, is a good time update this evidence. This protocol also addresses a knowledge gap relating to outcomes at discharge. This review will provide an important evidence synthesis that can add to the evaluation of safety of PR started during an inpatient admission. The protocol reads very well. I have just some minor suggestions to (in my view) help the rationale of this review and clarify some of the methods. Introduction: Section of text starting line 26 on page 6 and ending on line 8 of page 7: I think the authors have the most relevant articles cited here but I think the text would benefit from re-wording to reflect the interpretation of the evidence in these articles. For example, the ERS/ATS guideline (Wedzicha et al 2017) recommended that pulmonary rehabilitation be initiated within 3 weeks after hospital discharge and not to initiate pulmonary rehabilitation during hospitalisation. Indeed, the Puhan et al 2016 also concluded with evidence in favour of use of PR post-exacerbation. This manuscript (line 28, page 6) suggests that PR is recommended during and after AECOPD. The authors quite rightly go on to say there is controversy about safety during AECOPD. The uncertainty within the evidence synthesised by Puhan et al 2016 and the recommendation against PR during AECOPD in the ATS/ERS guideline is largely influenced by to one large trial (Greening et al 2014) where there was a negative effect of PR on mortality. However, the excess mortality was observed in the PR group only at 12 months, and so it is difficult to attribute the risk of mortality at this time-point directly to PR. The authors may consider highlighting this specific limitation and uncertainty (i.e. very low quality) in the evidence from Puhan et al and ATS/ERS guideline. I say this because it is provides a strong rationale for the evidence to be updated with the current review (i.e up-to-date synthesis to reduce uncertainty/increase the confidence in the evidence) and the particular focus on events/outcomes during AECOPD to better capture the safety of PR. Line 19, page 6: is it accurate to say 'permanent'? Would 'may be long-lasting without intervention' or something similar be more appropriate? Methods: PROSPERO: Please add registration number when available. Information sources: I notice you are searching until June 2020? Is that correct? If so, I would recommend updating your searches prior to submission of a completed review in the future. Search strategies:

	Not sure what is meant by "keyterm search terms" Data analysis: I wonder if the following sentence should be moved to the start of the section: "Patients who receive the usual care will be used as the control group in this study. Data will be synthesized by calculating mean differences and pooled odds ratios using random effects models in Review Manager 10.[14]" I say this because the MCIDs are more of an interpretation of the estimates from your random effect models. "The unit of analysis was the participant", should this be "will" Line 40-52, page 11: Could you consider adding some more information about how you will handle heterogeneity. For example, you mention 'sufficiently homogenous' and 'prohibited by heterogeneity'. It would be helpful to clarify what you will use to guide your decisions, I squared values?

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

1. In the Methods and Analysis; types of interventions - rehabilitation programmes may include self-management, education, dietary advice etc, how was this measured?

   AUTHORS' RESPONSE:
   While rehabilitation may include the elements of self-management, education, and dietary advice, etc. these are also part of standard acute care. These elements mentioned are important, but there has been little evidence that they provide a measurable effect on morbidity and mortality. Our focus is on the safety and efficacy of the aspects of rehabilitation related to exercise/mobilization/physical activity interventions that occur during the hospital stay.

   To clarify this, we have made the following changes. In the Introduction, we have revised it accordingly:

   Pulmonary rehabilitation (PR) during or shortly after an AECOPD may counteract these limitations in physical function. These programs focus on mobility, muscle strength, and/or exercise tolerance to mitigate the impact of the exacerbation and the hospitalization period.

2. The description of the time points was also slightly confusing as to the inclusion/ exclusion of studies.

   AUTHORS' RESPONSE:
   We think the reviewer is referring to this sentence: "…will be searched from their inception to June 2020…"

   We have revised this to now read:
   Bibliographic databases MEDLINE, Embase, the Cumulative Index to Nursing and Allied Health Literature (CINAHL), the Physiotherapy Evidence Database (PEDro), and trial registries

(Cochrane Controlled Register of Trials (CENTRAL), the US National Institutes of Health Ongoing Trials Register (ClinicalTrials.gov), the Allied and Complementary Medicine Database (AMED), PsycInfo and the World Health Organization (WHO) trials portal) will be searched up to June 2021 for potentially relevant articles using pre-defined search strategies (Appendix 1).

3. Data analysis describes the QoL and exercise capacity measures however this paper was to review the safety of inpatient PR as well as efficacy, and I was unable to find any description of how adverse events, readmission or LOS were analysed which are important in measuring the safety aspect. The authors may want to review the MID for the ISWT as this has now changed to 40m (Evans RA, Singh SJ. Minimum important difference of the incremental shuttle walk test distance in patients with COPD. Thorax 2019;74: 994-995)

AUTHORS' RESPONSE:
Thank you for this comment. We did state that we will collect information on adverse events. We have expanded Table 1, Outcomes as per the following:

Outcomes	• *Pre-intervention and post-intervention values of primary and secondary outcomes* • *Time points when data was collected* • *Adverse events related to the intervention, including but not limited to: reports of falls, increased length of stay, or responses to exercise (dizziness, chest pain, arrhythmia, hypo/hypertension, increased pain)*

Reviewer: 2

1. Introduction: Section of text starting line 26 on page 6 and ending on line 8 of page 7: I think the authors have the most relevant articles cited here but I think the text would benefit from re-wording to reflect the interpretation of the evidence in these articles. For example, the ERS/ATS guideline (Wedzicha et al 2017) recommended that pulmonary rehabilitation be initiated within 3 weeks after hospital discharge and not to initiate pulmonary rehabilitation during hospitalisation. Indeed, the Puhan et al 2016 also concluded with evidence in favour of use of PR post-exacerbation. This manuscript (line 28, page 6) suggests that PR is recommended during and after AECOPD. The authors quite rightly go on to say there is controversy about safety during AECOPD. The uncertainty within the evidence synthesised by Puhan et al 2016 and the recommendation against PR during AECOPD in the ATS/ERS guideline is largely influenced by to one large trial (Greening et al 2014) where there was a negative effect of PR on mortality. However, the excess mortality was observed in the PR group only at 12 months, and so it is difficult to attribute the risk of mortality at this time-point directly to PR. The authors may consider highlighting this specific limitation and uncertainty (i.e. very low quality) in the evidence from Puhan et al and ATS/ERS guideline. I say this because it is provides a strong rationale for the evidence to be updated with the current review (i.e up-to-date synthesis to reduce uncertainty/increase the confidence in the evidence) and the particular focus on events/outcomes during AECOPD to better capture the safety of PR.

AUTHORS' RESPONSE: Thank you for these thoughtful comments. We agree with your interpretation of both the guidelines and the studies which underpin the recommendations. We have revised the Introduction accordingly:

Frequent acute exacerbations are a common feature of chronic obstructive pulmonary disease (COPD), especially for patients with moderate to severe disease,[1]. Although most acute exacerbations of COPD (AECOPD) can be managed in the outpatient setting, many patients require hospital admission. The cost of these hospitalizations is immense, representing more than 25% of the total health care cost of COPD in the United States,[2]. Decreases in quality of life (QoL), lung function, quadriceps muscle force and physical activity have been reported in patients during and after hospitalization for AECOPD which, in many cases, may be permanent without intervention,[3]. Pitta et al,[4] reported that hospitalized patients with COPD show little improvement in their activity levels during their hospital stay, and by discharge only 9% of their day was spent standing or walking. Pulmonary rehabilitation (PR) during or shortly after an AECOPD may counteract these limitations in physical function. These programs focus on mobility, muscle strength, and/or exercise tolerance to mitigate the impact of the exacerbation and the hospitalization period.

A 2016 Cochrane systematic review by Puhan et al,[5] examined the effects of "early" PR (i.e. during or shortly after an AECOPD) on future hospital admissions and other patient-related outcomes such as mortality, health-related QoL, and exercise capacity in patients with COPD. However, this review included studies in which the PR intervention began in the hospital and was continued after discharge. The outcome measures assessed the effect of rehabilitation continued after discharge. It is possible that much of the benefit of PR during an AECOPD is seen during this post-discharge period. What is not clear is if a PR intervention delivered during the hospital admission is safe and results in changes in health status and physical function prior to discharge.

The Cochrane review authors concluded that early PR "improves quality of life and exercise capacity and is a safe intervention for patients with COPD after they have experienced an exacerbation",[5]. Despite this, however, the 2017 American Thoracic Society / European Respiratory Society guideline for the management of AECOPD provided a conditional recommendation against the initiation of PR during hospitalization, [6]. This caution was based on their own analysis that found that although PR initiated during hospitalization improved exercise capacity, it also increased mortality. The increased mortality finding was strongly influenced by one study,[7] which reported an increase in all-cause mortality in the intervention group. The excess mortality rate in the intervention group was detected at the 12-month post-discharge assessment, but did not occur during hospitalization, as there was no difference in hospital length of stay or mortality between the groups during the hospital period. Nevertheless, the question remains regarding the safety of rehabilitation specifically during this acute care, inpatient phase, when patients are most ill with an AECOPD,[6]. Clinicians who are considering implementing in-hospital PR for patients soon after admission must be confident when referring their patients to these programs. Therefore, the purpose of this systematic review is to evaluate the safety and efficacy of in-hospital PR for patients admitted to hospital for an AECOPD.

2. Line 19, page 6: is it accurate to say 'permanent'? Would 'may be long-lasting without intervention' or something similar be more appropriate?

AUTHORS' RESPONSE: Thank you for this comment. We have revised the manuscript to read: "*…may be permanent without intervention*".

3. Methods: PROSPERO: Please add registration number when available.

AUTHORS' RESPONSE: this has been added.

4. Information sources: I notice you are searching until June 2020? Is that correct? If so, I would recommend updating your searches prior to submission of a completed review in the future.

   AUTHORS' RESPONSE: Thank you for this comment. We have revised the methods to include publications up to June 2021.

5. Search strategies: Not sure what is meant by "keyterm search terms"

   AUTHORS' RESPONSE: Thank you for picking up this typo. We have corrected it to read: "keyword".

6. Data analysis: I wonder if the following sentence should be moved to the start of the section: "Patients who receive the usual care will be used as the control group in this study. Data will be synthesized by calculating mean differences and pooled odds ratios using random effects models in Review Manager 10.[14]" I say this because the MCIDs are more of an interpretation of the estimates from your random effect models.

   AUTHORS' RESPONSE: Great suggestion, we have done this.

7. "The unit of analysis was the participant", should this be "will"

   AUTHORS' RESPONSE: We have changed 'was' to 'is'.

8. Line 40-52, page 11: Could you consider adding some more information about how you will handle heterogeneity. For example, you mention 'sufficiently homogenous' and 'prohibited by heterogeneity'. It would be helpful to clarify what you will use to guide your decisions, I squared values?

   AUTHORS' RESPONSE: Thank you for this comment. The manuscript now reads:

   Data will be synthesized by calculating mean differences and pooled odds ratios using random-effects models in Review Manager 10,[9]. Heterogeneity will be assessed using forest plots, the Q statistic and the I^2 statistic. Heterogeneity will be examined using subgroup analysis and by analyzing items from the quality assessment that indicate high risk versus low risk of bias (for example, blinding during data collection procedures). High heterogeneity will be considered present if $p<0.1$ for the Q statistic or if I^2 is >50% ,[10]. If we find the included studies are sufficiently homogenous in terms of design, study population and outcomes, we will conduct a meta-analysis to calculate a pooled effect estimate. If prohibited by heterogeneity, a narrative synthesis will be conducted instead.

VERSION 2 – REVIEW

REVIEWER	Chaplin, Emma University Hospitals of Leicester NHS Trust, Pulmonary Rehabilitation
REVIEW RETURNED	25-May-2021

GENERAL COMMENTS	All comments have been answered comprehensively.

REVIEWER	Jones, Arwel Monash University, Department of Allergy, Immunology and Respiratory Medicine
REVIEW RETURNED	21-May-2021

GENERAL COMMENTS	Thank you for the comprehensive response to my previous comments. I wish you all the best with your review. I shall look forward to seeing your findings soon.